MATH3516: Discrete Mathematics

Unit 1: Fundamental Structures and Set Theory

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Contents

1 Formal Introduction to Discrete Mathematics **2**

2 Taxonomy of Discrete Mathematics **2**

3 Set Theory: Axiomatic Foundations **3**

4 Subsets and Inclusion **4**

5 Power Sets and Higher-Order Classes **4**

6 Set Operations and Boolean Algebra **4**

7 Venn Diagrams and Set Relations **5**

8 Finite Sets and Cardinality **5**

9 Principles of Combinatorial Counting **6**

10 The Cartesian Product **6**

1 Formal Introduction to Discrete Mathematics

Definition: Discrete Mathematics is the branch of mathematics dealing with objects that can assume only distinct, separated values. It focuses on *discrete structures* rather than continuous functions. Unlike calculus, which relies on the property of continuity ($\lim_{x \to a} f(x) = f(a)$), discrete mathematics operates on sets where elements possess no infinitesimal proximity to one another.

Computational Paradigm: In the context of Computer Science, the discrete nature of information is foundational. Digital architecture is predicated on binary states (bits), necessitating a mathematical framework that handles finite or countably infinite structures.

Illustrative Quantities in Computing:

1. **Quantized States:** The state space of an n-bit register is finite, defined by the set $\{0, 1\}^n$.
2. **Digital Resolution:** A display matrix of 1920×1080 represents a discrete coordinate system of $2,073,600$ individual addressable elements.
3. **Temporal Discretization:** CPU operations are governed by a discrete clock signal, where logic transitions occur only at specific intervals (cycles).

2 Taxonomy of Discrete Mathematics

The field is characterized by several interrelated disciplines:

- **Mathematical Logic:** The study of formal systems of deduction and the truth values of propositions.
- **Combinations:** The analysis of the number of ways certain patterns or structures can be formed.
- **Graph Theory:** The study of relational structures represented as vertices (nodes) and edges (links).
- **Number Theory:** An investigation into the properties of integers, specifically primality and divisibility.

Fundamental Example: The Division Algorithm

The most direct application of Number Theory in discrete systems is the decomposition of an integer into a quotient and a remainder.

Theorem: For any integer a (the dividend) and a positive integer d (the divisor), there exist unique integers q (the quotient) and r (the remainder) such that:

$$a = dq + r, \quad \text{where } 0 \leq r < d$$

Numerical Application: Let $a = 17$ and $d = 5$.

1. Performing the division, we find that 5 fits into 17 exactly 3 times with a remainder.

2. This gives us $q = 3$ and $r = 17 - (5 \times 3) = 2$.

3. **Verification:** $17 = 5(3) + 2$, where $0 \leq 2 < 5$.

Programming Context: In discrete computational structures, this theorem is implemented via:

- `q = a / d;` *(Integer Division)*
- `r = a % d;` *(Modulo Operation)*

Formal Applications:

1. **Permutations:** The cardinality of the set of all linear orderings of a set S with $|S| = 3$ is given by $P(3, 3) = 3! = 6$.

2. **Topological Connectivity:** In a complete graph K_n, the total number of edges is determined by the binomial coefficient $\binom{n}{2}$. For $n = 4$, $E = 6$.

3. **Fundamental Theorem of Arithmetic:** Every integer $n > 1$ has a unique factorization into primes. e.g., $60 = 2^2 \cdot 3 \cdot 5$.

Mathematical Derivation: The Binomial Coefficient

The expression $\binom{n}{k}$, known as the binomial coefficient, represents the number of ways to choose k elements from a set of n elements without regard to order.

General Formula:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Step-by-Step Calculation for $n = 4, k = 2$: To determine the number of edges (E) in a complete graph K_4:

1. **Substitute values:** $\binom{4}{2} = \frac{4!}{2!(4-2)!}$

2. **Expand factorials:** $\frac{4\times3\times2\times1}{(2\times1)\times(2\times1)}$

3. **Simplify numerator and denominator:** $\frac{24}{2\times2} = \frac{24}{4}$

4. **Result:** $E = 6$

Graph Theory Interpretation: In a network of 4 nodes, there are exactly 6 unique ways to form a connection between any two nodes.

3 Set Theory: Axiomatic Foundations

Definition: A set is a well-defined collection of distinct objects, considered as an object in its own right. The membership of an object x in a set S is denoted by $x \in S$.

Notational Convention: Sets are typically defined via the Roster Method (listing elements) or Set-Builder Notation (defining a characteristic property $P(x)$).

Mathematical Expressions:

1. **Finite Roster:** $A = \{10, 20, 30, 40\}$.
2. **Infinite Subset:** $\mathbb{P} = \{x \in \mathbb{Z}^+ \mid x \text{ is prime}\}$.
3. **Predicative Specification:** $C = \{x \in \mathbb{Z} \mid -2 \leq x \leq 2\} \implies C = \{-2, -1, 0, 1, 2\}$.

4 Subsets and Inclusion

Definition: A set A is a **subset** of B, denoted $A \subseteq B$, if and only if $\forall x(x \in A \implies x \in B)$. **Definition:** A is a **proper subset** ($A \subset B$) if $A \subseteq B \wedge A \neq B$.

Relational Examples:

1. Given $V = \{1, 2, 3, 4, 5\}$ and $E = \{2, 4\}$, the inclusion $E \subset V$ holds.
2. Reflexivity: For any set A, the statement $A \subseteq A$ is a tautology.
3. The empty set $\emptyset$ is a subset of every set S ($\emptyset \subseteq S$).

5 Power Sets and Higher-Order Classes

Definition (Class of Sets): A collection of sets is often termed a *family* or *class* to avoid Russell's Paradox in naive set theory.

Definition (Power Set): For any set S, the power set $\mathcal{P}(S)$ is the set of all subsets of S, defined as $\mathcal{P}(S) = \{A \mid A \subseteq S\}$. **Theorem:** If the cardinality $|S| = n$, then $|\mathcal{P}(S)| = 2^n$.

Cardinality Demonstrations:

1. **Set Families:** $\mathcal{F} = \{\{1, 5\}, \{10, 20\}, \{100\}\}$ is a family of sets of integers.
2. **Binary Power Set:** Let $S = \{1, 2\}$. $\mathcal{P}(S) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.
3. **Singleton Power Set:** Let $T = \{0\}$. $\mathcal{P}(T) = \{\emptyset, \{0\}\}$.

6 Set Operations and Boolean Algebra

The primary operations on sets include:

- **Union ($A \cup B$):** $\{x \mid x \in A \vee x \in B\}$.
- **Intersection ($A \cap B$):** $\{x \mid x \in A \wedge x \in B\}$.
- **Set Difference ($A \setminus B$):** $\{x \mid x \in A \wedge x \notin B\}$.

Operational Results:

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$.

1. $A \cup B = \{1, 2, 3, 4, 5, 6\}$.
2. $A \cap B = \{3, 4\}$.
3. $A \setminus B = \{1, 2\}$, whereas $B \setminus A = \{5, 6\}$.

7 Venn Diagrams and Set Relations

Venn diagrams serve as a heuristic tool for visualizing the logical relationship between sets within a Universal Set U.

Categorical Configurations:

1. **Non-empty Intersection:** When $A \cap B \neq \emptyset$, the regions overlap to indicate shared elements.
2. **Complementation:** The complement $A^c = U \setminus A$ represents all elements in the universe not contained in A.
3. **Disjoint Sets:** If $A \cap B = \emptyset$, the sets are mutually exclusive and represented by non-overlapping boundaries.

8 Finite Sets and Cardinality

Definition: A set S is **finite** if there exists a bijection between S and the set $\{1, 2, \ldots, n\}$ for some $n \in \mathbb{N}$. The integer n is the cardinality $|S|$.

Core Concept: The Bijective Function

A **bijection** (or bijective function) is a mapping between two sets that establishes a perfect one-to-one correspondence between their elements.

Formal Definition: A function $f : A \to B$ is a **bijection** if and only if it satisfies two conditions simultaneously:

1. **Injective (One-to-One):** Distinct elements in A map to distinct elements in B.

$$\forall x_1, x_2 \in A, (f(x_1) = f(x_2) \implies x_1 = x_2)$$

2. **Surjective (Onto):** Every element in B is mapped to by at least one element in A.

$$\forall y \in B, \exists x \in A \text{ such that } f(x) = y$$

Significance Bijections are the formal mechanism used to define *cardinality*. We state that two sets A and B have the same size ($|A| = |B|$) if and only if there exists a bijection between them.

Illustrative Example: Let $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$. The function $f = \{(a, 1), (b, 2), (c, 3)\}$ is a bijection. Each letter is paired with a unique number, and no numbers are left unassigned.

Cardinality Analysis:

1. The set $S = \{x \in \mathbb{Z}^+ \mid x \leq 100, 5|x\}$ has cardinality $|S| = 20$.
2. The set of solutions for $x^2 = 9$ in $\mathbb{Z}$ is $\{-3, 3\}$, thus $|A| = 2$.
3. The singleton set $\{\emptyset\}$ has cardinality 1, as it contains exactly one element (the empty set).

9 Principles of Combinatorial Counting

1. Sum Rule (Addition Principle): If a task can be done in n ways and a second task in m ways, and these tasks are mutually exclusive, there are $n + m$ ways to do one of the tasks.

2. Principle of Inclusion-Exclusion (PIE): To find the cardinality of the union of two finite sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Numerical Applications:

1. **Disjoint Selection:** Selecting one hardware component from 5 distinct vendors or 4 alternative vendors results in $5 + 4 = 9$ possibilities.
2. **PIE Application:** In a cohort of 35 students, where 20 are enrolled in Mathematics and 15 in Computer Science, with 5 enrolled in both, the total unique enrollment is $20 + 15 - 5 = 30$.
3. **Divisibility Analysis:** The number of integers $n \in [1, 100]$ such that $2|n \vee 5|n$ is calculated as: $\lfloor \frac{100}{2} \rfloor + \lfloor \frac{100}{5} \rfloor - \lfloor \frac{100}{10} \rfloor = 50 + 20 - 10 = 60$.

10 The Cartesian Product

Definition: The Cartesian product $A \times B$ is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$. Formally:

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

Product Examples:

1. If $A = \{1, 2\}$ and $B = \{3, 4\}$, then $A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$.
2. Discrete Plane: The set $\{0, 1\}^2$ represents the vertices of a unit square in a 2D discrete space.
3. If $|A| = m$ and $|B| = n$, then $|A \times B| = mn$.

Bibliography and Scholarly Resources

- *Discrete Mathematics and Its Applications*, Kenneth H. Rosen.
- *Concrete Mathematics: A Foundation for Computer Science*, Graham, Knuth, and Patashnik.